

HILBERT–BURCH RESOLUTIONS OF SECANT SUPPORTS ON ABELIAN FOURFOLDS

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ABSTRACT. We prove that the ideal sheaf of the support of a secant object on a principally polarised abelian fourfold has no Hilbert–Burch resolution of rank one, and none of rank two at the discriminants 1 and 5 when its Chern classes are polynomials in the polarisation, and we correct the route note that proposed these supports.

1. WHAT THIS NOTE DOES

Status 1.1. This note proves no case of the Hodge conjecture. It writes the secant route of [BMB, Section 18] on one principally polarised abelian fourfold as a list of steps, records which constructions are excluded, and proves one new exclusion, [Proposition 4.1](#): the Hilbert–Burch resolution of a secant support has rank $r \geq 2$, and $r \geq 3$ at $d = 1, 5$ when its Chern classes are polynomials in Θ . That proposition is also in the paper [BMB, Proposition 18.60]. The numerical statements of [Sections 5](#) and [6](#) are checked in exact arithmetic ([Section 8](#)); every other statement quoted here is proved in [BMB] or in the papers it cites, at the place given.

The target. Markman proved that the Weil classes are algebraic on the abelian sixfolds of split Weil type for every imaginary quadratic field [[Mar25a](#), [Mar25b](#)], and deduced it for all abelian fourfolds of Weil type, for every field and every discriminant, through a product argument of Schoen [[Sch98](#), Proposition 10]; the Hodge conjecture for abelian fourfolds follows [BMB, Section 1]. The fourfolds of Weil type are therefore not a target. The secant route starts from a principally polarised abelian fourfold X and a secant object F on it, and aims at the Weil classes $\mathbf{W}(K, 4, \delta)$ on the abelian eightfolds $X \times \hat{X}$ of Weil type, $K = \mathbb{Q}(\sqrt{-d})$ [BMB, Theorem 18.29(iv), Remark 18.69]. Besides an object on X it needs an affirmative answer to [[Mar25b](#), Question 11.4].

Where the route sits. The closure theorem [BMB, Theorem 20.58] lists five minimal sufficient sets of open statements,

$$\{\mathbf{F3}\}, \quad \{\mathbf{L}, \mathbf{M}\}, \quad \{\mathbf{L}, \mathbf{F3}'\}, \quad \{\mathbf{V}, \mathbf{F3}'\}, \quad \{\mathbf{P2}_{\text{split}}, \mathbf{F2}, \mathbf{F3}'\}.$$

Statement (F1) of the paper is the propagation statement, and it is needed only in the form $\mathbf{P2}_{\text{split}}$, for the split families of the CM fields of degree at least four: (P2) for an imaginary quadratic field and for a nontrivial discriminant is derived by descent [BMB, Proposition 20.13]. The secant route lies in no minimal sufficient set [BMB, Theorem 20.58(iv)]: granting both of its open demands gives $\mathbf{W}(K, n, \delta)$ for imaginary quadratic K and leaves the CM fields of degree at least four outside. Statement (F3), the conjecture for the varieties that are not abelian, is equivalent to the Hodge conjecture [BMB, Proposition 20.35]; (F3'), the conjecture modulo abelian varieties [BMB, Definition 20.36], is weaker; and (F2) is equivalent to the Hodge conjecture for abelian varieties [BMB, Proposition 21.49]. Every element of a minimal sufficient set other than $\mathbf{P2}_{\text{split}}$ is a consequence of the Hodge conjecture, and $\{\mathbf{F3}\}$ and $\{\mathbf{L}, \mathbf{M}\}$ are each equivalent to it [BMB, Theorem 20.58(vi)].

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2. THE REDUCTION TO ONE OBJECT

Let (X, Θ) be a principally polarised abelian fourfold, $d > 0$, and F a perfect complex on X with $\text{ch}(F) = a u_\Theta + b v_\Theta$, where u_Θ, v_Θ span the secant plane [BMB, Section 18]. No action of K on X is assumed: the field $K = \mathbb{Q}(\sqrt{-d})$ enters through $\text{ch}(F)$, and the Weil structure lives on $X \times \widehat{X}$ [BMB, Proposition 18.25(v), Theorem 18.29(iv)].

Theorem 2.1 ([BMB, Theorem 18.29(ii), (iii)]). *The classes of $HT^2(X)$ preserving $\text{ch}(F)$ to first order are the ten polarised first order deformations of (X, Θ) and the six Poisson directions compensated in $H^{0,2}$. The object F satisfies the weakened criterion if and only if*

(a) $v \lrcorner \text{At}(F) = 0$ in $\text{Ext}^2(F, F)$ for every $v \in H^1(T_X)$ with $v \lrcorner \Theta = 0$, that is, F extends to first order along every polarised deformation of (X, Θ) ; and

(b) $\text{At}(F)^2 = -d \Theta^2 \cdot \text{id}_F$ in $\text{Ext}^2(F, F \otimes \Omega_X^2)$.

The trace of (b) is automatic.

For Markman's object, built from translates of Θ , condition (a) holds and (b) is the whole condition [BMB, Corollary 18.30]. For any other candidate both are needed, and (a) runs over all ten polarised directions.

3. THE LOCAL DICHOTOMY AND THE NUMERICAL TARGETS

Let $F = \mathcal{I}_Z(b\Theta)$ with Z of codimension two. If the local ring $\mathcal{O}_{Z,p}$ is Cohen–Macaulay, then $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = 0$ for $q \geq 2$ [BMB, Theorem 18.48]. If Z fails to be Cohen–Macaulay at a point of one of the types of [BMB, Remark 18.49], then F fails the criterion; if Z is Cohen–Macaulay, the criterion is the vanishing of the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ for all $\theta, \theta' \in H^0(T_X)$, and when Z is smooth $\mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z) = N_{Z/X}$ [BMB, Corollary 18.50].

For a smooth support S , Grothendieck–Riemann–Roch with trivial Todd class forces [BMB, Theorem 18.40]

$$[S] = N\Theta^2, \quad N = \frac{1}{2}(b^2 + d), \quad \chi(\mathcal{O}_S) = (b^2 + d)(3b^2 - d),$$

$$K_S^2 = 3(b^2 + d)(7b^2 - d), \quad e(S) = 3(b^2 + d)(5b^2 - 3d), \quad K_S \cdot \lambda = 32bN, \quad \lambda^2 = 24N,$$

with $\lambda = \Theta|_S$, and the Bogomolov–Miyaoka–Yau inequality is exactly $d \leq b^2$. At $b = 3$ with d squarefree and N integral this leaves $d \in \{1, 3, 5, 7\}$ [BMB, Corollary 18.41], with the invariants of Table 1.

d	N	$\chi(\mathcal{O}_S)$	K_S^2	$e(S)$	$K_S \cdot \lambda$	λ^2
1	5	260	1860	1260	480	120
3	6	288	2160	1296	576	144
5	7	308	2436	1260	672	168
7	8	320	2688	1152	768	192

TABLE 1. The forced invariants of a smooth support at $b = 3$.

4. WHAT IS EXCLUDED

A candidate in any of the following classes fails before condition (b) is reached.

- (i) Complete intersections of two divisors [BMB, Theorem 18.54].
- (ii) Zero loci of sections of vector bundles of rank two, and more generally every Hilbert–Burch resolution of rank one (Proposition 4.1(i)).

- (iii) Cones of maps between sums of powers of an ample line bundle with class in $\mathbb{Q}\Theta$, whatever the rank [BMB, Theorem 18.63, Corollary 18.68(i)].
- (iv) Cones of maps between sums of simple isogeny blocks, that is, of simple semi-homogeneous bundles [BMB, Theorem 18.66, Remark 18.67]; these include all line bundles.
- (v) For a smooth support, resolutions whose two terms both have classes in the lattice generated by line bundles [BMB, Corollary 18.58].
- (vi) Markman’s explicit complex from N translates of Θ [BMB, Theorem 18.34].
- (vii) Hilbert–Burch resolutions of rank two at $d \equiv 1 \pmod{4}$, when the Chern classes are polynomials in Θ (Proposition 4.1(ii)).

Proposition 4.1 ([BMB, Proposition 18.60]). *Let (X, Θ) be a principally polarised abelian fourfold, $b \geq 1$, $d > 0$, and let $Z \subset X$ be of codimension two with a resolution $0 \rightarrow E_1 \rightarrow E_0 \rightarrow \mathcal{I}_Z \rightarrow 0$ by vector bundles of ranks r and $r+1$ and $\text{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + bv_\Theta$.*

- (i) *$r \geq 2$, whatever the bundles. Equivalently, Z is not the zero locus of a section of a vector bundle of rank two.*
- (ii) *If $r = 2$ and the Chern classes of E_1 are polynomials in Θ , then $3d - b^2 - 4ab \equiv 0 \pmod{12}$, where $c_1(E_1(b\Theta)) = a\Theta$. At $b = 3$ this says $d \equiv 3 \pmod{4}$.*

Proof. Put $E = E_1(b\Theta)$, $G = E_0(b\Theta)$, $F = \mathcal{I}_Z(b\Theta)$, so $c(G) = c(E)c(F)$, and Newton’s identities give

$$c(F) = 1 + b\Theta + N\Theta^2 + \frac{bN}{3}\Theta^3 + \frac{N(b^2-3d)}{12}\Theta^4.$$

A vector bundle of rank s has $c_k = 0$ for $k > s$. At $r = 1$, with $\ell = c_1(E)$, the rank two bundle G has $0 = c_3(G) = (\frac{bN}{3}\Theta + N\ell)\Theta^2$, so $\ell = -\frac{b}{3}\Theta$ by hard Lefschetz, and then $c_4(G) = -\frac{N(b^2+9d)}{36}\Theta^4 \neq 0$. A section of a rank two bundle V vanishing in codimension two on Z gives the Koszul resolution $0 \rightarrow \det V^\vee \rightarrow V^\vee \rightarrow \mathcal{I}_Z \rightarrow 0$, of rank one. At $r = 2$, with $c_1(E) = a\Theta$ and $c_2(E) = \frac{m}{2}\Theta^2$, the rank three bundle G has $0 = c_4(G) = c_4(F) + c_1(E)c_3(F) + c_2(E)c_2(F)$, which integrates to $6m = 3d - b^2 - 4ab$, and $\chi(E) = a^4 - 2a^2m + \frac{1}{2}m^2 \in \mathbb{Z}$ makes m even. $\square$

Remark 4.2. Neither condition restricts $r \geq 3$: a bundle of rank four has no vanishing Chern class on a fourfold, and at $r = 3$ the only condition, $c_4(E) = 0$, involves E alone [BMB, Remark 18.61]. From rank three on, Chern numbers alone decide nothing, and Corollary 6.2(iii) gives data at every $d \in \{1, 3, 5, 7\}$.

Status 4.3. Two shapes remain [BMB, Remark 18.69]: a smooth support at $d \in \{1, 3, 5, 7\}$ whose Hilbert–Burch resolution has rank $r \geq 2$, and $r \geq 3$ at $d = 1, 5$ in the polynomial case, with one term not projectively flat and not both terms in the lattice of line bundles; or a singular Cohen–Macaulay support with a resolution over a bundle that is not projectively flat. Neither is settled.

5. THE PROGRAM

Step 5.1 (Base point). Take any principally polarised abelian fourfold X ; d is fixed by the Chern character of the candidate, not by X . When X is simple, the injectivity of $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_S)$ required by [BMB, Lemma 18.62] holds for a smooth support automatically [BMB, Lemma 18.43]. When $\text{Hdg}^2(X) = \mathbb{Q}\Theta$ and $\text{Hdg}^4(X) = \mathbb{Q}\Theta^2$, as for a very general X , all Chern classes are polynomials in Θ and Proposition 4.1(ii) applies. An action of K on X is not needed, and it does not help with condition (a) (Step 5.3).

Step 5.2 (Support). Construct a smooth or Cohen–Macaulay surface $Z \subset X$ with $[Z] = N\Theta^2$ and, if smooth, the invariants of Table 1, with a Hilbert–Burch resolution $0 \rightarrow E \rightarrow G \rightarrow \mathcal{I}_Z(3\Theta) \rightarrow 0$ of rank $r \geq 2$ outside the classes of Section 4. The three families proposed in the first version of this note stand as follows.

- (1) Degeneracy loci of maps between Fourier–Mukai transforms. The transform of a nondegenerate line bundle on $\widehat{X}$ is a simple semi-homogeneous bundle [Muk78, Muk81], whose rank on a principally polarised fourfold is q^4 for slope $\frac{p}{q}\Theta$, never 2 or 3; a resolution with both terms sums of such bundles is excluded by [BMB, Theorem 18.66]. Transforms of other sheaves are not excluded.
- (2) Images of $C_1 \times C_2$ or of $C^{(2)}$. A smooth image $S \cong C_1 \times C_2$ has $K_S^2 = 8\chi(\mathcal{O}_S)$, which with Table 1 reads $5N = 4b^2$, impossible at $b = 3$ and possible only when $5 \mid b$. A smooth $S \cong C^{(2)}$ of genus g has $\chi(\mathcal{O}_S) = \frac{1}{2}(g-1)(g-2)$, never 260, 288, 308 or 320. Singular images must be Cohen–Macaulay and are not excluded.
- (3) Zero loci of sections of rank two bundles. Excluded by Proposition 4.1(i), for every bundle, equivariant or not.

Step 5.3 (Extension). Verify condition (a) of Theorem 2.1 along all ten polarised directions. If X carries an action of K of signature $(2, 2)$, the K -linear polarised directions form a space of dimension 4, so an equivariant construction that persists over the Weil family gives (a) along at most four of the ten directions; the other six destroy the action, and equivariance says nothing about them. For a smooth support, (a) is equivalent to the Bloch semiregularity map of S having rank six, the least value possible [BMB, Proposition 18.46].

Step 5.4 (The identity (b)). For a Cohen–Macaulay support, verify that the images of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ vanish for all $\theta, \theta' \in H^0(T_X)$ [BMB, Corollary 18.50(ii)]. The first version proposed computing these classes from the Atiyah classes of E and G restricted to Z ; that formula is a proposal, it is not in [BMB], and it has not been carried out for any candidate.

Step 5.5 (Propagation). If Steps 5.2 to 5.4 succeed for secant objects F_1, F_2 , then $\Phi(F_1 \boxtimes F_2)$ on $X \times \widehat{X}$ satisfies the weakened criterion [BMB, Theorem 18.29(iv)]. With an affirmative answer to [Mar25b, Question 11.4], the semiregularity argument of Buchweitz and Flenner [BF03] and Pridham [Pri24] would then make $\mathbf{W}(K, 4, \delta_0)$ algebraic on the eightfolds of the family, and [BMB, Proposition 20.13] every δ , for the discriminants reached and nothing beyond them [BMB, Remark 18.69].

6. THE NUMERICAL LEMMA, CORRECTED

Write the Chern classes of a class E of rank r as $c_k(E) = m_k \Theta^k / k!$; since $\Theta^k / k!$ is integral and primitive, integrality means $m_k \in \mathbb{Z}$, and m_4 is a number of points. Put $F = \mathcal{I}_Z(3\Theta)$ with $\text{ch } F = u_\Theta + 3v_\Theta$.

Lemma 6.1. *If $\text{ch } G = \text{ch } E + \text{ch } F$ with $\text{rk } G = r + 1$, then $c(G) = c(E)c(F)$, where $c(F)$ has $m_k(F) = (3, 2N, 6N, 6N(3-d))$; so G has integral Chern classes whenever E does. The formulas for $n_k = m_k(G)$ in the first version of this note are this product written out. With Bogomolov’s discriminant $\Delta(V) = 2 \text{rk } V c_2(V) - (\text{rk } V - 1)c_1(V)^2$, whose coefficient of Θ^2 is $r m_2 - (r-1)m_1^2$,*

$$\Delta(G) - \Delta(E) = (m_2 - (m_1 - 3)^2 + 18 + (r+1)d)\Theta^2.$$

The first version displayed $\text{rk } V c_2(V) - (\text{rk } V - 1)c_1(V)^2$, which differs from Bogomolov’s in the factor 2 on c_2 , while its values, such as $\Delta(E) = 4\Theta^2$ for $c(E) = 1 + \Theta^2$ in rank two, are Bogomolov’s. What the lemma does not see is the vanishing of $c_k(E)$ for $k > r$ and of $c_k(G)$ for $k > r + 1$, and this is the whole constraint at $r \leq 2$.

Corollary 6.2. *Let $b = 3$ and $d \in \{1, 3, 5, 7\}$, and suppose the Chern classes of E are polynomials in Θ .*

- (i) Rank two data exist exactly at $d = 3$ and $d = 7$. With $c_1(E) = a\Theta$ they have $\Delta(E) = (d + 1 - (a + 2)^2)\Theta^2$ and $\Delta(G) = \frac{1}{2}(9(d + 1) - 4a^2)\Theta^2$.
- (ii) The example of the first version, $c(E) = 1 + \Theta^2$ in rank two, has $c_4(G) = 180, 144, 84, 0$ points at $d = 1, 3, 5, 7$, so it is a resolution datum only at $d = 7$. With the conditions of the first version, $\Delta(E) > 0$, $\Delta(G) > 0$, $\mu(E) < \mu(G)$ and $\chi(E^\vee \otimes G) \geq 1$, rank two survives only at $d = 7$, for $a \in \{-2, -1, 0\}$; at $d = 3$ the two discriminants are positive only for $a \in \{-2, -1\}$, where $\chi(E^\vee \otimes G) = -78$ and -7 .
- (iii) In rank three, $c(E) = 1 + \Theta^2$ meets all four conditions at every $d \in \{1, 3, 5, 7\}$, with $\chi(E^\vee \otimes G) = 53, 101, 173, 269$.

Proof. At $r = 2$ the relation of [Proposition 4.1](#) gives $m_2 = \frac{1}{2}(d - 3) - 2a$, from which the two discriminants follow. The remaining values are finite computations, checked exactly in [Section 8](#); the closed form of $\chi(E^\vee \otimes G)$ at $r = 2$ is $\frac{1}{8}(24a^4 + 64a^3 - 88a^2d + 104a^2 - 64ad + 192a + 57d^2 - 174d + 153)$. $\square$

The four conditions of the first version are heuristics: Bogomolov’s inequality bounds Δ below for a stable bundle, and $\chi(E^\vee \otimes G) \geq 1$ neither implies nor is implied by the existence of a map $E \rightarrow G$. Chern numbers therefore decide ranks one and two and say nothing from rank three on.

7. THE TOY COMPUTATION

The first version searched unions of coordinate abelian surfaces $\{x_i = c_i, x_j = c_j\}$ in $X = E_1 \times E_2 \times E_3 \times E_4$ with the product polarisation, at $b = 1$, for configurations with the Chern character of a secant support, Cohen–Macaulay everywhere, and with a combinatorial condition proposed there as necessary for (a). The programs are in `toy/`. A rerun of the search at $b = 1$, $d = 1$ with forty restarts of 60000 steps found an exact configuration on four fibres per factor, twelve surfaces with the required pair and triple counts and $\chi(\mathcal{O}_Z) = 4$, and none on three fibres per factor, where the best configuration missed by one in a triple count. At $d = 3$ twenty restarts on each size found none; the best configurations missed the targets by a total of 31 and 16. The symbolic powers of the union of the six coordinate surfaces through a point give no admissible Chern character for $1 \leq m \leq 20$.

This says nothing about [Step 5.2](#) on a simple fourfold: $b = 1$ is not Markman’s normalisation, X is not simple and $\text{NS}(X)$ has rank at least four, so neither [\[BMB, Lemma 18.43\]](#) nor [Proposition 4.1\(ii\)](#) applies, and abelian surfaces do not survive a deformation of X to a simple fourfold.

8. THE CHECKS

The program `route_checks.py` performs nineteen checks in exact arithmetic: the invariants of [Table 1](#) and the class $[S] = N\Theta^2$; [Lemma 6.1](#) and the formulas of the first version on random data; the Bogomolov normalisation; the example and the three parts of [Corollary 6.2](#); the value $c_4(G) = -6N(1 + d)$ points at $r = 1$, $b = 3$; the conditions on products and symmetric squares of curves; the rank q^4 of a simple semi-homogeneous bundle; and the dimension pq of the K -linear polarised directions for signature (p, q) . Item (LXX) of [\[BMB, Appendix D\]](#) checks [Proposition 4.1](#).

9. CHANGES FROM THE FIRST VERSION

- (1) The target. The Hodge conjecture for abelian fourfolds of Weil type is Markman’s theorem; the route aims at $\mathbf{W}(K, 4, \delta)$ on eightfolds and needs [\[Mar25b, Question 11.4\]](#).

- (2) The closure framing. (F1) is needed only as $P2_{\text{split}}$, for CM fields of degree at least four; the secant route lies in no minimal sufficient set; [BMB, Proposition 20.35] concerns (F3), not (F3').
- (3) The base point needs no K -action, and a K -action gives condition (a) along at most four of ten directions.
- (4) The class of a smooth support is $N\Theta^2$, not $N\Theta^2/2$.
- (5) The local statement is one implication, and the global condition is a family of classes indexed by pairs θ, θ' .
- (6) Family (3) is excluded, family (1) is excluded for transforms of line bundles, and smooth products and symmetric squares of curves are excluded at $b = 3$.
- (7) The numerical lemma omits the vanishing above the rank; its example works only at $d = 7$; its displayed discriminant lacks a factor 2.
- (8) The bibliography is that of [BMB]: [Mar25b] is “Secant sheaves and Weil classes on abelian varieties”, and Pridham’s paper is [Pri24].

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